

# Joint Labor Search and the Taxation of Couples

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– Work in progress –

# Motivation I: Joint Labor Search

- **Families** coordinate their labor supply.
- **Frictional labor markets**: families coordinate *job search strategy*.
  - How hard to search on and off the job (i.e., endog. search intensity).
  - Which offers to accept (i.e., reservation wages).
  - When to quit endogenously (e.g., after some kind of family shock).
- Papers on **joint search** have explored these margins.
  - “Breadwinner cycle” (Guler et al., 2012), gender gaps (Flabbi & Mabli, 2018), marital wage premium (Pilossoph & Wee, 2021)
- We combine **joint search** of couples with **assets** and **precautionary job search** (Chaumont/Shi, 2022). The composition of couples matters, also for firms.

## Motivation II: Policy

- Many countries: **Unit of taxation** is the household (e.g., Germany, U.S.)
- With progressive taxes, joint income taxation imposes **high marginal and participation tax rates on secondary earners**
- On the other hand, **lower marginal tax rates for primary earners**
- **Unemployment insurance** (UI) typically individual-based, while other transfers are often means-tested (partner's income, assets)
- Previous literature: Frictionless labor markets, exogenous wages
- **We ask:** how are couples' **job-search strategies, their job mobility and earnings dynamics shaped by the tax-transfer system?**

# This Paper

- Study couples' job search behavior using the U.S. SIPP (1996-2005)
- Build **random-search equilibrium model** with single and couple households
- We abstract from gender(!), inactivity vs. active search, marriage and divorce, home production, children (for now?)
- Explore effects of tax-transfer policies (individual- vs. household-based) for employment, labor market dynamics, earnings, savings, and welfare.  
→ Today: tax experiment, move from joint to individual taxation.

## What is new?

|                                             |   |                        |
|---------------------------------------------|---|------------------------|
| Job-ladder model, endogenous search efforts | } | 3 insurance mechanisms |
| Workers risk averse, save in riskless asset |   |                        |
| Couples: intrahousehold risk sharing        |   |                        |

GE effects: Firms post vacancies, match-specific productivity, wages

# Related Literature

## **Joint vs. separate taxation**

Guner, Kaygusuz and Ventura (2012); Olivetti and Petrongolo (2017); Borella, de Nardi and Yang (2023); Bick and Fuchs-Schündeln (2018); Holter, Krueger and Stepanchuk (2023); Bierbrauer, Boyer, Peichl and Weishaar (2023)

## **Household joint decision-making in search-matching models**

Guler, Guvenen and Violante (2012); Flabbi and Mabli (2018); Dey and Flinn (2008); Pilossoph and Wee (2021); Holzner and Schulz (2023), Bacher et al. (2025)

## **Secondary earners in models with endogenous labor demand**

Mankart and Oikonomou (2017); Birinci (2021)

## **The marriage market matters and the transmission of policies**

Gayle and Shephard (2019); Foerster et al. (2024); Almar et al. (2025)

# EMPIRICS

# Data

- Survey of Income and Program Participation (SIPP), 1996-2005 waves
- Sample: Individuals aged 25-55, married (spouse present) and never married
- Drop self-employed, disabled, permanently inactive, students, retirees, etc.
- 55,096 unique obs, 1.3M person-months
- Median duration in panel is 36 months
- Average flows between employment (E) and non-employment (N) for married/single, net of compositional differences (age, edu, gender, kids)

## Married (spouse present)

- NE: 7.11% per month
- EE: 1.64% per month
- EN: 1.02% per month

## Singles (never married)

- NE: 12.68% per month
- EE: 1.94% per month
- EN: 1.42% per month

# Some facts about couples in the SIPP

- Survey data, we follow a regression-based approach:
  - Idea: regress outcome of interest (e.g., the EE rate) on observables and a dummy that distinguishes couple types and individuals (e.g., one-earner/two-earner, primary/secondary earner).
  - Report marginal effects of dummy, evaluated at the mean.
  - This approach nets out compositional differences across couple types in terms of the included observables.
  - This is the heterogeneity that we do not want include in our model (e.g., age, education, marriage market sorting, kids).
  - Easy to take into account survey weights, etc.
  - Same approach to generate calibration targets below.
- Plan: Log hourly wages, net liquid wealth, labor market transition rates

# Log hourly wages across couple types

- Employed individuals in one-earner couples earn more ( $\approx 10\%$ ).

|                          | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                |
|--------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Two-earner couple        | 4.060<br>(0.00392) | 4.079<br>(0.00375) | 4.075<br>(0.00375) | 4.072<br>(0.00335) | 4.069<br>(0.00334) | 4.096<br>(0.00392) |
| One-earner couple        | 4.221<br>(0.00932) | 4.112<br>(0.00942) | 4.133<br>(0.00933) | 4.151<br>(0.00781) | 4.168<br>(0.00776) | 4.192<br>(0.00985) |
| Time Controls            | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Gender                   | No                 | Yes                | Yes                | Yes                | Yes                | Yes                |
| No. of kids              | No                 | No                 | Yes                | Yes                | Yes                | Yes                |
| Own Controls (age & edu) | No                 | No                 | No                 | Yes                | Yes                | Yes                |
| Own and Partner Controls | No                 | No                 | No                 | No                 | Yes                | Yes                |
| Net liquid wealth        | No                 | No                 | No                 | No                 | No                 | Yes                |
| Observations             | 760113             | 760113             | 760113             | 760113             | 760113             | 182487             |

# Log hourly wages by primary earner status

- Large and significant difference within two-earner households ( $\approx 60\%$ ).

|                              | (1)                | (2)                | (3)                | (4)                | (5)                | (6)                |
|------------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Two-earner couple, secondary | 3.826<br>(0.00483) | 3.843<br>(0.00489) | 3.842<br>(0.00487) | 3.859<br>(0.00451) | 3.847<br>(0.00443) | 3.875<br>(0.00552) |
| Two-earner couple, primary   | 4.363<br>(0.00431) | 4.341<br>(0.00460) | 4.342<br>(0.00459) | 4.321<br>(0.00401) | 4.336<br>(0.00401) | 4.354<br>(0.00485) |
| Time Controls                | Yes                | Yes                | Yes                | Yes                | Yes                | Yes                |
| Gender                       | No                 | Yes                | Yes                | Yes                | Yes                | Yes                |
| No. of kids                  | No                 | No                 | Yes                | Yes                | Yes                | Yes                |
| Own Controls (age & edu)     | No                 | No                 | No                 | Yes                | Yes                | Yes                |
| Own and Partner Controls     | No                 | No                 | No                 | No                 | Yes                | Yes                |
| Net liquid wealth            | No                 | No                 | No                 | No                 | No                 | Yes                |
| Observations                 | 645659             | 645659             | 645659             | 645659             | 645659             | 156641             |

# Net liquid wealth across couple types

- One-earner couples are richer in terms of net liquid wealth ( $\approx$  USD 2.5K).

|                          | (1)                | (2)                | (3)                | (4)                | (5)                |
|--------------------------|--------------------|--------------------|--------------------|--------------------|--------------------|
| Two-earner couple        | 11983.1<br>(157.7) | 12041.2<br>(159.6) | 11948.0<br>(158.0) | 11959.5<br>(151.3) | 11844.8<br>(148.3) |
| One-earner couple        | 13512.9<br>(452.7) | 13168.1<br>(455.0) | 13720.7<br>(460.5) | 13652.6<br>(429.7) | 14332.7<br>(433.5) |
| Time Controls            | Yes                | Yes                | Yes                | Yes                | Yes                |
| Gender                   | No                 | Yes                | Yes                | Yes                | Yes                |
| No. of kids              | No                 | No                 | Yes                | Yes                | Yes                |
| Own Controls (age & edu) | No                 | No                 | No                 | Yes                | Yes                |
| Own and Partner Controls | No                 | No                 | No                 | No                 | Yes                |
| Observations             | 188103             | 188103             | 188103             | 188103             | 188103             |

## Job-to-job mobility by couple type

- No difference conditional on demographics, but significant difference conditional on wealth (0.16pp)

|                          | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|--------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Two-earner couple        | 0.0150<br>(0.000192) | 0.0150<br>(0.000193) | 0.0151<br>(0.000193) | 0.0151<br>(0.000193) | 0.0152<br>(0.000194) | 0.0120<br>(0.000302) |
| One-earner couple        | 0.0160<br>(0.000443) | 0.0159<br>(0.000457) | 0.0157<br>(0.000459) | 0.0154<br>(0.000458) | 0.0152<br>(0.000461) | 0.0136<br>(0.000814) |
| Time Controls            | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Gender                   | No                   | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| No. of kids              | No                   | No                   | Yes                  | Yes                  | Yes                  | Yes                  |
| Own Controls (age & edu) | No                   | No                   | No                   | Yes                  | Yes                  | Yes                  |
| Own and Partner Controls | No                   | No                   | No                   | No                   | Yes                  | Yes                  |
| Net liquid wealth        | No                   | No                   | No                   | No                   | No                   | Yes                  |
| Observations             | 723346               | 723346               | 723346               | 723346               | 723346               | 173323               |

# Job-to-job mobility by primary earner status

- Significant difference within two-earner couples ( $\approx 0.2\text{pp}$ ), sign flips conditional on wage

|                              | (1)                  | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
|------------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
| Two-earner couple, secondary | 0.0156<br>(0.000253) | 0.0159<br>(0.000268) | 0.0159<br>(0.000267) | 0.0158<br>(0.000266) | 0.0159<br>(0.000268) | 0.0138<br>(0.000262) |
| Two-earner couple, primary   | 0.0142<br>(0.000269) | 0.0139<br>(0.000289) | 0.0139<br>(0.000289) | 0.0140<br>(0.000290) | 0.0139<br>(0.000292) | 0.0162<br>(0.000321) |
| Time Controls                | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| Gender                       | No                   | Yes                  | Yes                  | Yes                  | Yes                  | Yes                  |
| No. of kids                  | No                   | No                   | Yes                  | Yes                  | Yes                  | Yes                  |
| Own Controls (age & edu)     | No                   | No                   | No                   | Yes                  | Yes                  | Yes                  |
| Own and Partner Controls     | No                   | No                   | No                   | No                   | Yes                  | Yes                  |
| Wage                         | No                   | No                   | No                   | No                   | No                   | Yes                  |
| Observations                 | 612313               | 612313               | 612313               | 612313               | 612313               | 588759               |

# Wage cuts

- Now conditioning EE rates on no wage cut
- 46% for primary earner, 59% for secondary earner
- Bigger gap remains (0.3pp, 32%)

|                              | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   |
|------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Two-earner couple, secondary | 0.00908<br>(0.000179) | 0.00937<br>(0.000192) | 0.00938<br>(0.000192) | 0.00934<br>(0.000192) | 0.00941<br>(0.000194) |
| Two-earner couple, primary   | 0.00680<br>(0.000176) | 0.00641<br>(0.000191) | 0.00641<br>(0.000190) | 0.00646<br>(0.000191) | 0.00636<br>(0.000193) |
| Time Controls                | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| Gender                       | No                    | Yes                   | Yes                   | Yes                   | Yes                   |
| No. of kids                  | No                    | No                    | Yes                   | Yes                   | Yes                   |
| Own Controls (age & edu)     | No                    | No                    | No                    | Yes                   | Yes                   |
| Own and Partner Controls     | No                    | No                    | No                    | No                    | Yes                   |
| Observations                 | 607898                | 607898                | 607898                | 607898                | 607898                |

# Taking Stock

## Wages:

- Employed individuals in one-earner couples earn more ( $\approx 10\%$ )
- Large and significant difference within two-earner households ( $\approx 60\%$ )

## Wealth:

- One-earner couples are richer in terms of net liquid wealth ( $\approx$  USD 2.5K)

## Job Mobility:

- Difference in job mobility when comparing employed individuals in one-earner and two-earner households depends on wealth.
- Relatively large differences within two-earner households, driven by wages.
- EE transitions with wage cuts are important.

# MODEL

# Households

- Continuous time
- Infinitely-lived two-worker households, discount rate  $\rho$
- Workers can be employed or non-employed
- Risk averse, disutility from job search  $2u(c/2) - k(s_1) - k(s_2)$ ,
- Can save and borrow in risk-free bond with interest rate  $r$

# Firms and Matching

- Create vacant jobs at flow cost  $K$ , free entry
- Upon meeting, worker-job pair draws match-specific output  $y \sim H(\cdot)$
- Firms are risk neutral, discount rate  $\rho_f$
- Random matching of vacant jobs and workers,
  - Non-employed worker with search effort  $s$  meets job at Poisson rate  $sp(\theta)$
  - Employed worker meets job at rate  $\bar{x}sp(\theta)$
  - Vacant jobs meets worker at rate  $q(\theta) = \lambda(\theta)/\theta$
  - Tightness  $\theta$  is the ratio between vacancies and aggregate search effort
  - Godfather shocks hit at rate  $\chi$  (Bagger & Lentz, 2019)
- Exogenous separation rate  $\delta$ , quit opportunity rate  $\gamma$

# Wages

- Assume that worker and firm bargain over flow wage at every instant, where breakdown leads to loss of income at same instant (Elsby and Gottfries 2022)
- Specifically, firm receives zero profits and worker gets  $z$  in case of breakdown
- Then flow bargaining leads to wage sharing rule

$$w(y) \equiv \eta y + (1 - \eta)z$$

if  $y \geq z$ , where  $\eta$  is the worker's relative bargaining power

- $z$  can be home production, minimum wage, etc. (special case:  $z = 0$ )
- Matches with  $y < z$  not viable
- Distribution of productivity upon meeting  $F(y)$

# Government

- Household gross income  $Y$  subject to tax-transfer schedule, where net income is

$$N(Y) \equiv \tau_0 Y^{1-\tau_1}$$

- Parameters  $\tau_0$  and  $\tau_1$  characterize level and progressivity respectively
- Workers entering non-employment are entitled to UI benefit  $\rightarrow b(y)$
- Depend on last wage which is monotonically linked to match productivity
- We set  $b(y) = \psi y$  with  $\psi$  a replacement rate
- UI benefits expire at Poisson rate  $\zeta$
- Subsistence level  $\bar{f}$  (food stamps, housing vouchers, etc.)

► Government budget constraint

# HJB Equation

Couple with two employed workers:

► Single-earner HH

► No-earner HH

► Distribution

$$\begin{aligned} \rho v^{ee}(w_1, w_2, a) = & \max_{c, s_1, s_2} 2u(c/2) - k(s_1) - k(s_2) \\ & + \bar{x}s_1 p(\theta) \int_y \max [v^{ee}(w(y'_1), w_2, a) - v^{ee}(w_1, w_2, a), 0] dH(y'_1) \\ & + \bar{x}s_2 p(\theta) \int_y \max [v^{ee}(w_1, w(y'_2), a) - v^{ee}(w_1, w_2, a), 0] dH(y'_2) \\ & + \chi p(\theta) \int_y [v^{ee}(w(y'_1), w_2, a) - v^{ee}(w_1, w_2, a)] dH(y'_1) \\ & + \chi p(\theta) \int_y [v^{ee}(w_1, w(y'_2), a) - v^{ee}(w_1, w_2, a)] dH(y'_2) \\ & + \delta [v^{ne}(b(y_1), w_2, a) - v^{ee}(w_1, w_2, a)] + \delta [v^{en}(w_1, b(y_2), a) - v^{ee}(w_1, w_2, a)] \\ & + \gamma \max [0, v^{ne}(b(y_1), w_2, a) - v^{ee}(w_1, w_2, a)] \\ & + \gamma \max [0, v^{en}(w_1, b(y_2), a) - v^{ee}(w_1, w_2, a)] \\ & + v_a^{ee} [ra - c + \max [N(w_1 + w_2), \bar{f}]] \end{aligned}$$

# Firm Value

## Employee's spouse is also employed

Employee's spouse is non-employed

Job Creation

$$\begin{aligned}
 (\rho_f + \delta + \gamma_1^* + \mu_1^* + \chi p(\theta)) J^{ee}(y_1, w_2, a) &= y_1 - w(y_1) \\
 &+ \bar{x} s_2^* p(\theta) \int_y \mathbb{1}_{2,EE}^*(y_2') [J^{ee}(y_1, w(y_2'), a) - J^{ee}(y_1, w_2, a), 0] dH(y_2') \\
 &+ \chi p(\theta) \int_y [J^{ee}(y_1, w(y_2'), a) - J^{ee}(y_1, w_2, a)] dH(y_2') \\
 &+ (\delta + \gamma_2^*) [J^{en}(y_1, b(w_2), a) - J^{ee}(y_1, w_2, a)] \\
 &+ J_a^{ee} [ra - c^* + \max [N(w(y_1) + w_2), \bar{f}]]
 \end{aligned}$$

- Firm hiring employee (wlog spouse 1) in a job with output  $y_1$ .
- HH's value is  $v^{ee}(w(y_1), w_2, a)$ . Optimal consumption is  $c^* = c^*(w(y_1), w_2, a)$ .
- $\gamma_i^* = \gamma_i^*(y_1, w_2, a)$  is spouse  $i$  quitting policy.  $s_2^*$  is spouse 2 optimal search effort.  
 $\mathbb{1}_{2,EE}^*(y_2') = \mathbb{1}(w_1(y_1), w_2, a, w_2(y_2'))$  is spouse 2 accept. rule, job offer prod.  $y_2'$
- $\mu_1^* = \mu_1^*(y_1, w_2, a, \theta)$  search and new job acceptance rate of the employee.

# CALIBRATION AND POLICY EXPERIMENT

# Calibration: External

- Monthly frequency calibration, setting:
  - $\rho = 0.0034$  to match the 0.96 discount rate annually
  - $r = 0.0009$  to match annual 1% risk-free rate
  - $\zeta = 0.1823$  to match the expected UI duration of 6 months
  - $\eta = 0.5$  - equal weight bargaining over wage
  - $\psi = 0.4$  to get 40% replacement rate in UI
- We set  $k(s) = \frac{s^2}{2}$
- We take the utility function to be  $u(c) = \frac{c^{1-\sigma}}{1-\sigma}$  and put  $\sigma = 2$
- No borrowing:  $\underline{a} = 0$
- Tax function:  $\theta_0 = 0.94$  and  $\theta_1 = 0.16$  (Holter et al. (2019))

# Calibration: Internal

- We fix the rate at which opportunist to quit arrive  $\gamma = 1$  (termination notice) and set  $\delta$  to match the EN rate of 1.02% per month
- $p(\theta)$  matches the non-employment share of 12.5%
- Matching function  $M(S, V) = A\sqrt{SV}$ , tightness  $\theta = V/S$ ,  $\theta$  picked to deliver  $q(\theta)$  job fill. rate of 0.7,  $A$  adjusted to match  $p(\theta)$
- $\bar{f}$  matches 10% replacement rate of average wage
- $\bar{x}$  matches the EE rate of 1.64% given  $p(\theta)$
- $H(y)$  is truncated log-normal with parameters  $(\mu_y, \sigma_y^2)$  and we set  $\underline{y}$  and  $\bar{y}$  as the 1<sup>st</sup> and 99<sup>th</sup> percentiles;  $\mu_z$  matches avg. taxable income normalisation to 1;  $\sigma_y^2$  is set to match the mean-min ratio of  $\approx 1.5$
- $\chi$  matches the 0.7% EE rate with wage cut of primary earners in double E households

# Internally Calibrated Parameters

| $\delta$ | $p(\theta)$ | $\bar{f}$ | $\bar{x}$ | $\sigma_y$ | $G$   | $K$  | $A$   | $\mu_y$ | $\chi$ |
|----------|-------------|-----------|-----------|------------|-------|------|-------|---------|--------|
| 0.102    | 0.035       | 0.056     | 1.51      | 0.239      | 0.049 | 5.53 | 0.157 | -0.279  | 0.260  |

- We set 30 grid points for wages, 30 for UI, 1 for UI expired, 50 nodes for assets; that gives approx. 190,000 distinct states for a household
- We hit all targets with accuracy of 1% of their value

# Model Fit

|                            | Data | Model |                        | Data | Model |
|----------------------------|------|-------|------------------------|------|-------|
| Employment                 | 87.5 | 87.5  | EN rate <sup>†</sup>   | 1.02 | 1.02  |
| Nonemployment <sup>†</sup> | 12.5 | 12.5  | One earner             | 1.30 | 1.02  |
| Single-earner couples      | 25.2 | 22.8  | Dual, primary          | 0.60 | 1.02  |
| Dual-earner couples        | 73.8 | 76.0  | Dual, secondary        | 1.29 | 1.02  |
| No-earner couples          | 0.9  | 1.2   |                        |      |       |
| EE rate <sup>†</sup>       | 1.64 | 1.65  | Avg. wage <sup>‡</sup> |      |       |
| One earner                 | 1.53 | 1.33  | One earner             | 1.01 | 1.01  |
| Dual, primary              | 1.48 | 1.16  | Dual, primary          | 1.04 | 1.11  |
| Dual, secondary            | 1.75 | 2.12  | Dual, secondary        | 0.97 | 0.88  |
| NE rate                    | 7.01 | 7.01  |                        |      |       |
| Zero earners               | –    | 9.81  |                        |      |       |
| One earner                 | –    | 6.12  |                        |      |       |

<sup>†</sup> Calibration target. <sup>‡</sup> Relative to overall average wage.

# Steady State Additional Characteristics

| Category                   | Search Effort | Marginal Tax Rate |
|----------------------------|---------------|-------------------|
| Employment <sup>†</sup>    |               |                   |
| One earner                 | 0.316         | 0.145             |
| Dual-earner (primary)      | 0.149         | 0.223             |
| Dual-earner (secondary)    | 0.385         | 0.223             |
| Nonemployment <sup>†</sup> |               | —                 |
| Zero earners               | 2.78          | —                 |
| One earner                 | 1.90          | —                 |

<sup>†</sup> In percent.

# From Joint to Individual Taxation

Experiment: decrease progressivity from 0.16 to 0.11 (that of singles in current US tax system)  
+ adjust tax level to balance the budget.

| Outcome                                | Change vs. Baseline |
|----------------------------------------|---------------------|
| <i>Individual Progressive Taxation</i> |                     |
| Non-employment                         | +0.58 pp            |
| Job Offer Arrival Rate                 | - 10.13 %           |
| Avg. Productivity                      | -0.26 %             |
| Avg. Consumption                       | -0.59 %             |
| Avg. Search Effort                     | +6.00 %             |
| Avg. Assets                            | +22.7 %             |
| Welfare                                | -2.57 %             |

Changes relative to joint taxation baseline.

# Details

- Lower progressivity  $\rightarrow$  less public insurance (redistribution)  $\rightarrow$  more self-insurance (savings)  $\rightarrow$  heterogeneous effects on different HH types  $\rightarrow$  some households search *less* (due to wealth effect), others *more* (precautionary job search dominates)
- Separate taxation  $\rightarrow$  lower marginal tax rates for second earners: employees and non-employed in single-earner hhds search *more*

| HH type/worker             | NN    | EN <sub>e</sub> | EN <sub>n</sub> | EE <sub>prim</sub> | EE <sub>sec</sub> |
|----------------------------|-------|-----------------|-----------------|--------------------|-------------------|
| $\Delta$ Search effort (%) | -3.03 | +1.09           | +6.52           | +1.31              | +2.27             |

- Agg. search effort composition effect  $\rightarrow$  firms are more likely to meet with employees (who reject less productive offers + leave jobs faster)  $\rightarrow$  lower job creation  $\rightarrow$  negative equilibrium effect of the policy reform

# Decomposing change in Non-Employment Rate

- Composition of households changes:  
NN 1.2%  $\rightarrow$  1.4%, EN 22.8%  $\rightarrow$  23.5%, EE 76%  $\rightarrow$  75.1%
- We take the job offer arrival rate  $p(\theta)$ , search effort functions and consumption functions from the pre-reform equilibrium and replace them one by one with those from the post-tax reform equilibrium

| Reform Effect          | Total | Savings | Labor Demand | Search Effort |
|------------------------|-------|---------|--------------|---------------|
| $\Delta N$ rate (p.p.) | +0.58 | +0.47   | +0.95        | -0.72         |

- Complex interactions: lower firm entry  $\rightarrow$  lower assets, higher search effort; higher search effort  $\rightarrow$  lower match duration  $\rightarrow$  lower filled job value

# PRELIMINARY CONCLUSIONS

# Conclusions

- There important differences in labor market transition rates and job-to-job mobility across couple types. Interactions with income (wages) and net wealth.
- We develop a new model of joint job search with assets and endogeneous labor demand that can match these differences.
- Three different insurance mechanisms (couple, search effort, assets)
- In this setting, changing the tax-transfer system leads to complex interactions and GE effects (through the firm side)
- Some married individuals search more, others will save more. Higher churn in the labor market discourages job creation. Overall, slightly negative welfare effect.

# SUPPLEMENTARY MATERIAL

# Transitions into N by couple type

- No significant difference.

|                          | (1)                   | (2)                   | (3)                   | (4)                   | (5)                   | (6)                   |
|--------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| Two-earner couple        | 0.00947<br>(0.000150) | 0.00912<br>(0.000146) | 0.00921<br>(0.000148) | 0.00931<br>(0.000149) | 0.00932<br>(0.000150) | 0.00798<br>(0.000249) |
| One-earner couple        | 0.00884<br>(0.000338) | 0.0108<br>(0.000357)  | 0.0103<br>(0.000359)  | 0.00974<br>(0.000354) | 0.00971<br>(0.000357) | 0.00770<br>(0.000640) |
| Time Controls            | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| Gender                   | No                    | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |
| No. of kids              | No                    | No                    | Yes                   | Yes                   | Yes                   | Yes                   |
| Own Controls (age & edu) | No                    | No                    | No                    | Yes                   | Yes                   | Yes                   |
| Own and Partner Controls | No                    | No                    | No                    | No                    | Yes                   | Yes                   |
| Net liquid wealth        | No                    | No                    | No                    | No                    | No                    | Yes                   |
| Observations             | 730258                | 730258                | 730258                | 730258                | 730258                | 174734                |

# Transitions into N by primary earner status

- Secondary earners are 80% more likely to transition into non-employment.

|                          | (1) | (2) | (3) | (4) | (5)                   | (6)                   | (7)                   | (8)                   |
|--------------------------|-----|-----|-----|-----|-----------------------|-----------------------|-----------------------|-----------------------|
| primary_earner2=0        |     |     |     |     | 0.0117<br>(0.000224)  | 0.0100<br>(0.000380)  | 0.00898<br>(0.000203) | 0.00758<br>(0.000349) |
| primary_earner2=1        |     |     |     |     | 0.00653<br>(0.000196) | 0.00549<br>(0.000341) | 0.00899<br>(0.000233) | 0.00806<br>(0.000414) |
| Time Controls            |     |     |     |     | Yes                   | Yes                   | Yes                   | Yes                   |
| Gender                   |     |     |     |     | Yes                   | Yes                   | Yes                   | Yes                   |
| No. of kids              |     |     |     |     | Yes                   | Yes                   | Yes                   | Yes                   |
| Own Controls (age & edu) |     |     |     |     | Yes                   | Yes                   | Yes                   | Yes                   |
| Own and Partner Controls |     |     |     |     | Yes                   | Yes                   | Yes                   | Yes                   |
| Net liquid wealth        |     |     |     |     | No                    | Yes                   | No                    | Yes                   |
| Hourly Wage              |     |     |     |     | No                    | No                    | Yes                   | Yes                   |
| Observations             |     |     |     |     | 618213                | 149586                | 594096                | 145501                |

# Government Budget Constraint

Exogenous spending  $G$ . Let household-level income from work be:

$$W^{ee} = w_1 + w_2, \quad W^{en} = w_1, \quad W^{ne} = w_2, \quad W^{nn} = 0,$$

and total transfers to a household be:

$$B^{ee} = 0, \quad B^{en} = b_2, \quad B^{ne} = b_1, \quad B^{nn} = b_1 + b_2,$$

Then, the government budget constraint reads:

$$\sum_{d \in \{nn, en, ne, ee\}} \int \left( Y_{\text{tax}}^d(x) - D^d(x) \right) h^d(x) d\Xi^d = G,$$

where:

$$Y_{\text{tax}}^d(x) = W^d(x), \quad Y^d(x) = W^d(x) + B^d(x), \quad D^d(x) = \max \left\{ N(Y^d(x)), \bar{f} \right\}.$$

# HJB Equation

Single-earner household (symmetry for  $en$  and  $ne$ ): [Go back](#)

$$\begin{aligned}\rho v^{ne}(b_1, w_2, a) &= \max_{c, s_1, s_2} 2u(c/2) - k(s_1) - k(s_2) \\ &+ s_1 p(\theta) \int_y \max [v^{ee}(w(y'_1), w_2, a) - v^{ne}(b_1, w_2, a), 0] dH(y'_1) \\ &+ \bar{x} s_2 p(\theta) \int_y \max [v^{ne}(b_1, w(y'_2), a) - v^{ne}(b_1, w_2, a), 0] dH(y'_2) \\ &+ \chi p(\theta) \int_y [v^{ee}(w_1, w(y'_2), a) - v^{ee}(w_1, w_2, a)] dH(y'_2) \\ &+ \zeta [v^{ne}(0, w_2, a) - v^{ne}(b_1, w_2, a)] \\ &+ \delta [v^{nn}(b_1, b(w_2), a) - v^{ne}(b_1, w_2, a)] \\ &+ \gamma \max [0, v^{nn}(b_1, b(w_2), a) - v^{ne}(b_1, w_2, a)] \\ &+ v_a^{ne} [ra - c + \max [N(b_1 + w_2), \bar{f}]]\end{aligned}$$

Note the notation is wlog as when  $b_1 = 0$  the UI expiry effect is nil.

# HJB Equation

Double-non-employed household: [▶ Go back](#)

$$\begin{aligned}\rho v^{nn}(b_1, b_2, a) &= \max_{c, s_1, s_2} 2u(c/2) - k(s_1) - k(s_2) \\ &+ s_1 p(\theta) \int_y \max [v^{en}(w(y'_1), b_2, a) - v^{nn}(b_1, b_2, a), 0] dH(y'_1) \\ &+ s_2 p(\theta) \int_y \max [v^{ne}(b_1, w(y'_2), a) - v^{nn}(b_1, b_2, a), 0] dH(y'_2) \\ &+ \zeta [v^{nn}(0, b_2, a) - v^{nn}(b_1, b_2, a)] \\ &+ \zeta [v^{nn}(b_1, 0, a) - v^{nn}(b_1, b_2, a)] \\ &+ v_a^{nn} [ra - c + \max [N(b_1 + b_2), \bar{f}]]\end{aligned}$$

# Firm Value

Employee's spouse is non-employed [Go back](#)

$$\begin{aligned}(\rho_f + \delta + \gamma_1^* + \mu_1^* + \chi p(\theta)) J^{en}(y_1, b_2, a) &= y_1 - w(y_1) \\ &+ s_2^* p(\theta) \int_y \mathbb{1}_{2,NE}^* [J^{ee}(y_1, w(y'_2), a) - J^{en}(w_1, b_2, a), 0] dH(y'_2) \\ &+ \zeta [J^{en}(y_1, 0, a) - J^{en}(y_1, b_2, a)] \\ &+ J_a^{en} [ra - c^* + \max [N(w(y_1) + b_2), \bar{f}]]\end{aligned}$$

# Job Creation

Given total search effort

$$S = \sum_{d \in \{nn, en, ne, ee\}} \int [s_1^{*,d}(x) + s_2^{*,d}(x)] h^d(x) d\Xi^d,$$

market tightness  $\theta = V/S$ , worker contact rate per unit search  $p(\theta)$ , and vacancy contact rate  $q(\theta) = p(\theta)/\theta$ , free entry implies

$$K = \frac{q(\theta)}{S} \sum_{d \in \{nn, en, ne, ee\}} \int \sum_{i \in \{1,2\}} m_i^d(x) s_i^{*,d}(x) \underbrace{\int \mathbb{1}_i^{*,d}(x, y) J^{d \rightarrow d_i^+}(y; x) dH(y)}_{\equiv \mathcal{J}_i^d(x)} \times h^d(x) d\Xi^d$$

- $x$  collects the state variables in  $d$  (e.g.,  $x = (w_1, w_2, a)$  etc.).
- Meeting efficiency:  $m_i^d(x) = \mathbf{1}\{i \text{ is non-employed in } d\} + \bar{x} \mathbf{1}\{i \text{ is employed in } d\}$ .
- Post-hire state  $d_i^+$ :  $nn \xrightarrow{i=1} en, nn \xrightarrow{i=2} ne; en \xrightarrow{i=2} ee; ne \xrightarrow{i=1} ee; ee \xrightarrow{i} ee$ .
- $\mathbb{1}_i^{*,d}(x, y)$  is the optimal acceptance rule;  $J^{d \rightarrow d_i^+}(y; x) \in \{J^{en}, J^{ne}, J^{ee}\}$  is the firm value at hiring.

# Distribution of Households

- **State-specific densities (given tightness  $\theta$ )**

$$h^{ee}(w_1, w_2, a), h^{en}(w_1, b_2, a), h^{ne}(b_1, w_2, a), h^{nn}(b_1, b_2, a),$$

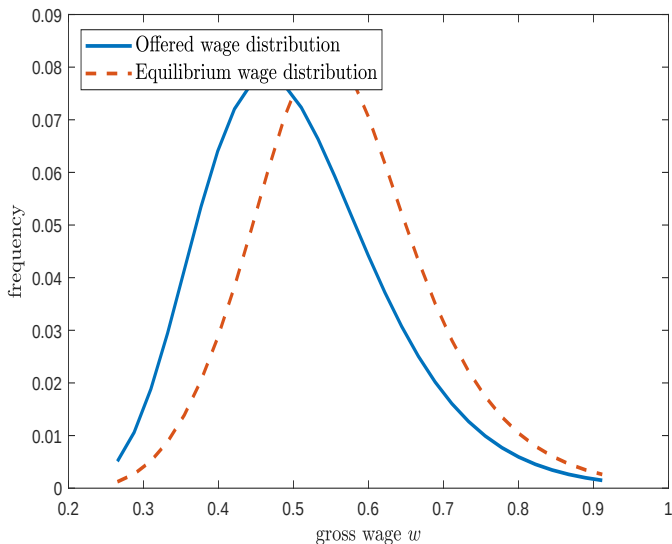
where the measure elements are  $d\Xi^{ee} = dw_1 dw_2 da$ ,  $d\Xi^{en} = dw_1 db_2 da$ , etc.

Normalisation:

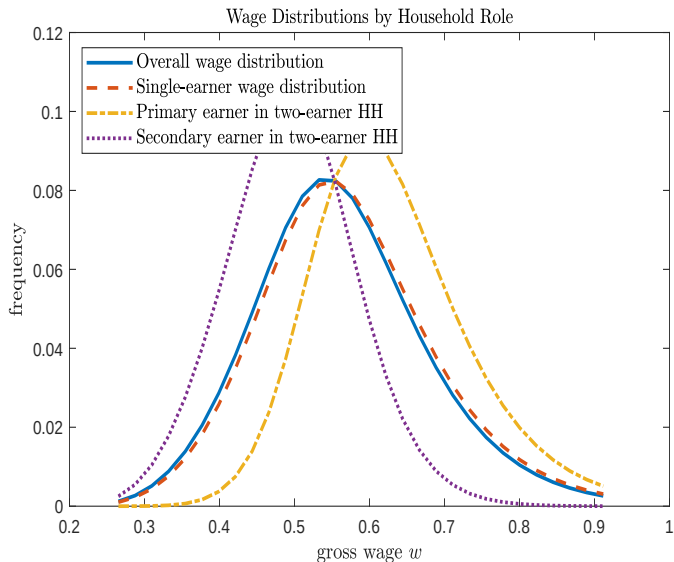
$$\int h^{ee} d\Xi^{ee} + \int h^{en} d\Xi^{en} + \int h^{ne} d\Xi^{ne} + \int h^{nn} d\Xi^{nn} = 1.$$

- **Stationarity (forward Kolmogorov).**  $\text{inflow}^s(\cdot) = \text{outflow}^s(\cdot)$  for each  $s \in \{ee, en, ne, nn\}$ , with flows generated by separations  $\delta$ , UI expiry  $\zeta$ , optimal quits  $\gamma_i^*$ , optimal search  $s_i^*$ , acceptance policies, forced relocations  $\psi p(\theta)$ , and the asset drift  $\dot{a}_t = ra_t - c_t^* + \max\{N(w_1 + w_2), \bar{f}\}$ .

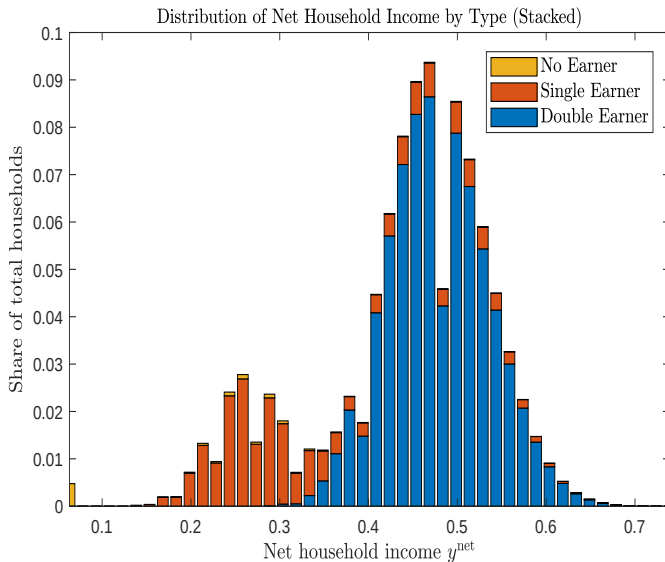
# Wage Offer and Equilibrium Distributions



# Wage Distributions by Household Role



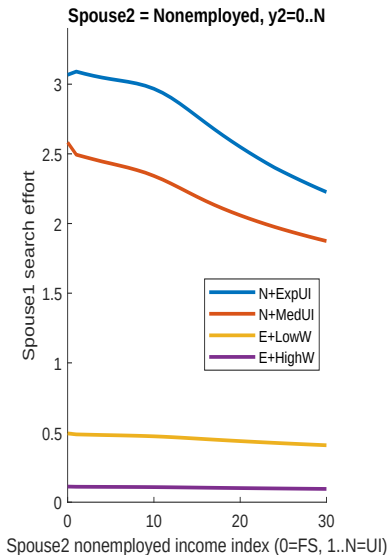
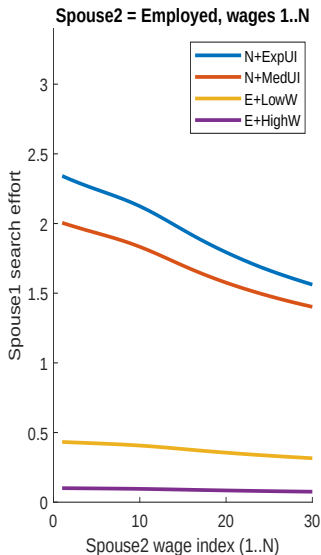
# Net Household Income Distribution



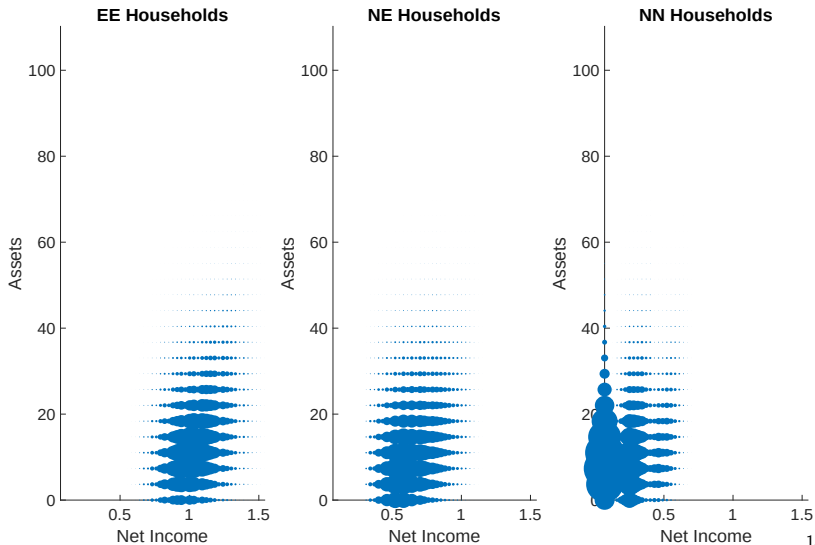
# Joint Search?

- We pick four workers: one with expired UI, one receiving moderate UI, one earning a low wage, and one with a high wage
- We integrate assets out, and calculate average search effort as a function of spousal labour market status and income
- Own search effort decreases in own and spousal income (the latter effect weaker the higher my own income)
- Self-insurance through search: when my spouse is non-employed and receives UI, I search harder than if their UI has already expired

# Average Search Effort



# HH Distributions: Assets and Income



# Firm Values

